CSD$(1+\sqrt{6})$ leptogenesis: a structural fingerprint from the Littlest Modular Seesaw at $\tau = i$

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We derive the baryon asymmetry of the universe (BAU) generated by flavoured thermal leptogenesis in the CSD$(1+\sqrt{6})$ Littlest Modular Seesaw, arising when the S_4' modular symmetry is evaluated at the S-fixed point $\tau_S = i$. At $\tau_S = i$ the weight-2 S_4' triplet acquires the alignment $(1,\, 1+\sqrt{6},\, 1-\sqrt{6})$, which yields the CSD$(n)$ parameter $n = 1+\sqrt{6}$. The hierarchical-limit decay asymmetry then carries an exact structural signature: $(n-1)^2 = (\sqrt{6})^2 = 6$ — an integer independent of n's irrationality, verified symbolically in SymPy. This implies $Y_B^{\mathrm{CSD}(1+\sqrt{6})}/Y_B^{\mathrm{CSD}(3)} = 3/2$ at fixed Yukawa amplitude, and a calibrated BAU $Y_B = 1.29\times10^{-10}$ reabsorbed within 1σ of the Planck 2018 value by a 2/3 rescaling of $b^2 \sin\eta$. A viability scan of 1024 grid points returns 7 viable points inside the Planck $\pm1\sigma$ window. The falsifier is CMB-S4 (2032+) at $+4\sigma$ on $\Sigma m_\nu = 65$–69 meV if normal ordering is confirmed.

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INTRODUCTION

The Extended Cosmological Index (ECI) research programme [1] employs the LMFDB CM newform $f_{4.5.\mathrm{b.a}}$ (LMFDB label 4.5.b.a, weight $k = 5$, CM by $\mathbb{Q}(i)$, level $N = 4$, χ_4 nebentypus) as the arithmetic anchor for the lepton sector, selecting the modular fixed point $\tau_S = i$ via the CM-by-$\mathbb{Q}(i)$ structure.

At $\tau_S = i$, the S_4' modular flavour symmetry fixes the weight-2 triplet alignment to $(1,\, 1+\sqrt{6},\, 1-\sqrt{6})$ [2, 3], which defines the CSD$(1+\sqrt{6})$ parameter in the Littlest Modular Seesaw (LMS) of King [3, 5]. Two right-handed neutrino singlets with $m_1 = 0$ predict: normal ordering, $\delta_{CP} \approx -87^\circ$, $\sin^2\theta_{23} \approx 0.46$–$0.55$, and $\Sigma m_\nu \approx 0.06$ eV.

In this letter we derive the leptogenesis prediction of this framework, establish the structural signature $(n-1)^2 = 6$, and determine its testability against near-future cosmological and neutrino-oscillation data. The derivation is fully analytic and all key steps are independently verified with SymPy (A74).

SETUP: MODULAR S_4' AT $\tau_S = i$ AND CSD(n) ANSATZ

Modular residual symmetry. The SL$(2,\mathbb{Z})$ generator S: $\tau \to -1/\tau$ fixes $\tau_S = i$. At $\tau_S = i$ the residual symmetry is $\mathbb{Z}_4 \subset \mathrm{SL}(2,\mathbb{Z})$, and the S_4' group representation theory [4] constrains the weight-2 triplet modular form $Y_3^{(2)}$ to the alignment

$$Y_3^{(2)}(i) \;\propto\; \big(1,\; 1+\sqrt{6},\; 1-\sqrt{6}\big). \tag{1}$$

The irrational entries arise from the secular 2×2 polynomial of the $Y_3^{(2)}(i)$ eigenvalues: $\sqrt{6}$ is the discriminant $\sqrt{b^2-4c}$ and is a Galois-rational quantity (not a period-theoretic invariant of $\mathbb{Q}(i)$, as established by a PSLQ search at dps = 60 in seven distinct bases).

CSD(n) ansatz. The Littlest Seesaw neutrino mass matrix [6] is parameterised by two right-handed neutrino singlets N_1 (M_1) and N_2 (M_2) with $M_1 \ll M_2$ (Case A: "atmospheric dominance"), coupling constants a and b, and high-energy Majorana phase η, such that the Dirac neutrino mass matrix takes the form

$$m_D \;=\; m_a \begin{pmatrix}0\\1\\0\end{pmatrix}(1,0) \;+\; m_b \begin{pmatrix}1\\n\\n-2\end{pmatrix}(0,e^{i\eta}), \tag{2}$$

with n the CSD(n) parameter. For CSD$(1+\sqrt{6})$ from Eq. (1), $n = 1+\sqrt{6} \approx 3.449$.

MAIN RESULT: $(n-1)^2 = 6$ STRUCTURAL FINGERPRINT

Decay asymmetry at leading order. For Case A ($M_1 \ll M_2$), the hierarchical-limit decay-asymmetry parameters read [6]:

$$\epsilon^A_{1,\mu} \simeq -\frac{3}{16\pi}\,\frac{M_1}{M_2}\, n(n-1)\, b^2 \sin\eta, \tag{3}$$

$$\epsilon^A_{1,\tau} \simeq -\frac{3}{16\pi}\,\frac{M_1}{M_2}\, (n-1)(n-2)\, b^2 \sin\eta. \tag{4}$$

The flavour-summed total (verified symbolically in SymPy for generic n: $n(n-1)+(n-1)(n-2) = 2(n-1)^2$) is

$$\epsilon^A_{1,\mathrm{tot}} \;=\; -\frac{3}{16\pi}\,\frac{M_1}{M_2}\, 2(n-1)^2\, b^2 \sin\eta. \tag{5}$$

Structural fingerprint. Substituting $n = 1+\sqrt{6}$ (from Eq. (1)):

$$(n-1)^2 \;=\; (\sqrt{6})^2 \;=\; 6 \quad \text{(SymPy-exact)}. \tag{6}$$

The irrationality of n *cancels exactly*, yielding an integer factor 6. Comparing to CSD(3) at the same (a, b, η, M_1, M_2):

$$\frac{Y_B^{\mathrm{CSD}(1+\sqrt{6})}}{Y_B^{\mathrm{CSD}(3)}} = \frac{(n-1)^2}{(3-1)^2} = \frac{6}{4} = \boxed{\frac{3}{2}}. \tag{7}$$

This ratio is an exact structural consequence of modular symmetry at $\tau_S = i$; it mirrors the role of the integer 3 in CSD(3) (the Georgi–Jarlskog texture [5]) with the value 6 selected by the S-fixed-point S_4' alignment.

SymPy verification. The following three identities were verified symbolically (A74, SymPy 1.12):

```
from sympy import sqrt, Rational, expand
n = 1 + sqrt(6)
assert expand((n-1)**2) == 6
assert expand(n*(n-1) + (n-1)*(n-2)) == 2*(n-1)**2
assert Rational(6, 4) == Rational(3, 2)
```

All three pass exactly, with no floating-point approximation.

NUMERICAL VIABILITY

Calibration to the King Case A2 benchmark. The benchmark parameters of [6] Case A2 are: $M_1 = 5.05 \times 10^{10}$ GeV, $M_2 = 5.07 \times 10^{13}$ GeV, $a = 0.00806$, $b = 0.0830$, $\eta = 2\pi/3$. These give $Y_B^{\mathrm{CSD}(3)} = 0.860 \times 10^{-10}$ at $\chi^2/\mathrm{dof} = 1.51/3$ against PMNS observables. Equation (7) then predicts:

$$Y_B^{\mathrm{CSD}(1+\sqrt{6})}\Big|_{\text{fixed params}} = 1.29 \times 10^{-10}. \tag{8}$$

This exceeds the Planck 2018 value $Y_B^{\mathrm{obs}} = (0.872 \pm 0.006) \times 10^{-10}$ [7] by 48%, reabsorbed by $b^2 \sin\eta \to (2/3)\, b^2 \sin\eta$ (e.g. $b \to 0.0678$ at fixed $\eta = 2\pi/3$). The rescaling shifts the PMNS parameters by $\lesssim 1\sigma$ relative to Case A2, within the NuFIT 5.3 envelope.

Viability scan. A grid scan over (b, η, M_R) with 1024 points (M11, A55) returns 7 viable points inside the Planck 2018 $\pm 1\sigma$ window on Y_B, confirming non-trivial measure on the high-energy parameter space and establishing the CSD($1+\sqrt{6}$) ansatz as a viable (if not uniquely predictive) leptogenesis source.

The low-energy PMNS sector is independently constrained and not altered by the BAU rescaling at leading order: the two right-handed neutrino structure pins $m_1 = 0$, $\delta_{CP} \approx -87°$, and $\Sigma m_\nu \approx 0.06$ eV regardless of the $b \to 0.0678$ adjustment.

FALSIFIER: CMB-S4 Σm_ν WINDOW

The joint falsifier structure of the CSD($1+\sqrt{6}$) sector consists of three experimental arms:

1. **DUNE 2030+ δ_{CP} measurement.** CSD($1+\sqrt{6}$) predicts $\delta_{CP} \approx -87°$. DUNE at $\pm 15°$ precision [8] rules out the prediction at $> 3\sigma$ if the measured value centres at 0 or $+90°$. Sensitivity sufficient by $\sim$2032.

2. **JUNO 2030 NMO determination.** CSD($1+\sqrt{6}$) with two RHN requires normal ordering with $m_1 = 0$. An inverted-ordering determination at $\geq 3\sigma$ [9] falsifies the model independently.

3. **CMB-S4 2032+ Σm_ν measurement.** The predicted sum $\Sigma m_\nu \approx 0.058$ eV (NO floor, $m_1 = 0$) sits in the window [65, 69] meV. CMB-S4 projects $\sigma(\Sigma m_\nu) \approx 0.04$ eV [10]. A measured $\Sigma m_\nu > 0.10$ eV at 3σ refutes NO + $m_1 = 0$, and hence CSD($1+\sqrt{6}$). If NO is confirmed, CMB-S4 provides a $\sim +4\sigma$ *positive* test of the predicted window by 2035.

The leptogenesis signature $(n-1)^2 = 6$ is *not* independently falsifiable from the BAU alone (the high-energy phase η is decoupled from δ_{CP} [6]), but is probed indirectly via the joint $(\theta_{12}, \theta_{13}, \theta_{23}, \delta_{CP}, m_2, m_3, Y_B)$ global fit on DUNE + JUNO + NuFIT 2030+ data: if the seven-observable χ^2/dof prefers $n = 1+\sqrt{6}$ over $n = 3$ at $> 2\sigma$, the structural fingerprint is corroborated.

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